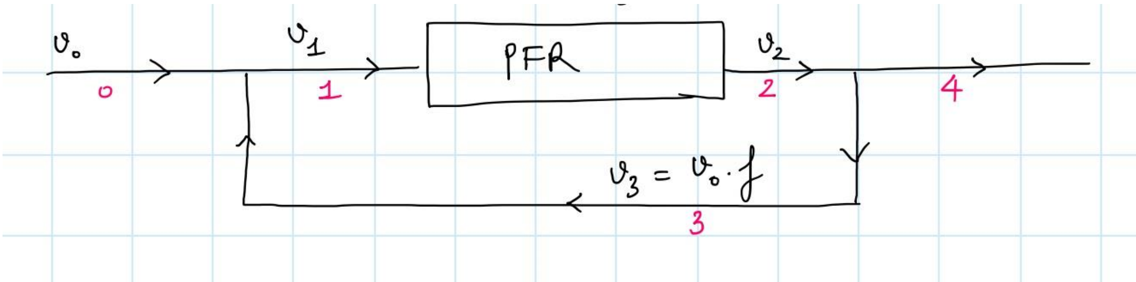
1. (50 points) For a first order reaction ($A \xrightarrow{k} B$) in a PFR volume V, a fraction f (f constant) of the exit volumetric flow rate is fed back to the PFR. Express the total conversion in term of f, k,V,v_0



2. (50 points) The exothermic reaction A→ B+C was carried out adiabatically and the following data recorded.

X	0	0.2	0.4	0.45	0.5	0.6	0.8	0.9
$-r_A$ (mol/L.min)	1	1.67	5	5	5	5	1.25	0.91

The entering molar flow rate of A was 300 mol/min.

(a) What are the PFR and CSTR volumes necessary to achieve 40% conversion?

(b) Over what range of conversion would the CSTR and PFR reactor volumes be identical?